Strand 4: Handling Data

Sub-strand 2: Chance and Probability

Lesson 1: Probability, Sample Spaces, and Dependent Events

1. Curriculum Standard and Overview

- Strand: 4 — Handling Data
- Sub-strand: 2 — Chance and Probability
- Content Standard: B9.4.2.1
- Indicators:
- B9.4.2.1.1 — Perform a probability experiment involving two dependent events (without replacement).
- B9.4.2.1.2 — Express probabilities as fractions, decimals, percentages, and ratios.
- Mock-paper links: P2 Q5(a) (pens in a bag: red, blue, green; probability as fraction/decimal/percentage/ratio; two pens without replacement).

2. Preamble — Chance in Everyday Language

"The chance of rain is 30%." "A coin has a 1 in 2 chance of heads."

Probability puts a number on how likely an event is:

$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}$$

The result is always between 0 and 1 (or 0% and 100%):

- 0 = impossible
- 1 = certain

3. Basic Vocabulary

Word	Meaning
Experiment	an action with an uncertain result (toss a coin, pick a pen)
Outcome	one possible result
Sample space S	the set of all possible outcomes
Event	a collection of outcomes we care about

4. Probability in Different Forms

Mock P2 Q5(a): A bag has 5 red, 3 blue, 2 green pens. Total = 10.

1. P(red) = 5/10 = ½
2. P(blue) = 3/10

- (α) decimal: 0.3
- (β) percentage: 30%
- (γ) ratio: 3 : 7 (favourable : not-favourable) or 3 : 10 (blue : total) — in BECE, write 3 : 7 when asked "as a ratio" of the event to its complement, or 3 : 10 if they mean blue to total. State which you mean.

Always simplify fractions.

5. Two-Stage Experiments

5.1 With replacement (independent)

The bag is unchanged between picks. Multiply the same probabilities.

5.2 Without replacement (dependent) — Mock P2 Q5(a)(iii)

If 2 pens are picked in succession without replacement, P(both green):

1. First pick: green pens = 2, total = 10 → P(G_(1)) = 2/10 = ⅕.
2. Second pick: green pens now = 1, total = 9 → P(G_(2) mid G_(1)) = 1/9.
3. Both green: ⅕ × 1/9 = 1/45.

Answer: 1/45.

Key idea: without replacement, the denominator and the numerator change after the first draw.

6. Tree Diagrams (Preview)

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G (2/10) → G (1/9) → P(GG) = 2/10 × 1/9 = 1/45
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not G (8/10) → ...
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Along each branch multiply; across branches add.

7. Twenty Fully Worked Examples

Example 1 — Mock P2 Q5(a)(i)

P(red) = 5/10 = ½.

Example 2 — Mock P2 Q5(a)(ii)

P(blue) = 3/10 = 0.3 = 30% = 3 : 7 (or 3 : 10).

Example 3 — Mock P2 Q5(a)(iii)

P(both green) = 1/45.

Example 4 — Coin toss

P(heads) = ½.

Example 5 — Fair die

P(3) = ⅙; P(even) = 3/6 = ½.

Example 6 — Certain event

P(a day has 24 hours) = 1.

Example 7 — Impossible event

P(rolling a 7 on a die) = 0.

Example 8 — Complement

P(not red) = 1 - ½ = ½.

Example 9 — Cards

P(ace) = 4/52 = 1/13.

Example 10 — Two coins (with replacement)

P(both heads) = ½ × ½ = ¼.

Example 11 — Without replacement (letters)

Bag: A, A, B. Pick two without replacement. P(both A) = ⅔ × ½ = ⅓.

Example 12 — Percentage form

P = ⅜ = 0.375 = 37.5%.

Example 13 — Ratio form

P(win) = 2/7. Odds in favour = 2 : 5.

Example 14 — Spinner

Spinner with 4 equal sectors, 1 red: P(red) = ¼.

Example 15 — At least one

Die thrown twice. P(at least one six) = 1 - P(no six) = 1 - ≤ft(⅚)^(2) = 11/36.

Example 16 — Mixed bag

Bag: 5 red, 3 blue, 2 green. P(not green) = 8/10 = ⅘.

Example 17 — Two without replacement (red then blue)

5/10 × 3/9 = ⅙.

Example 18 — Two without replacement (blue then red)

3/10 × 5/9 = ⅙.

Example 19 — Order matters in calculation

P(red then blue) = P(blue then red) in this bag, but the working differs.

Example 20 — Word problem

20 pupils, 8 girls. P(girl) = 8/20 = ⅖ = 0.4 = 40%.

8. Comprehensive Exercise Sets (40 Questions)

The authentic mock probability set (P2 Q5(a)) is included in full. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Basic probability (includes mock)

1. [Mock 2026 P2 Q5(a)(i)] In a pen bag, 5 are red, 3 are blue and 2 are green. What is the probability of picking a red pen?

2. What is P(blue) in the same bag?
3. What is P(green) in the same bag?
4. A fair coin is tossed. Find P(heads).
5. Spiral review: The mean of 5, 8, 4, 7 and 2x is 6. Find x.

Exercise 2 — Different forms (includes mock)

1. [Mock 2026 P2 Q5(a)(ii)] Express the probability of picking a blue pen as a
(α) decimal; (β) percentage; (γ) ratio.
2. Express ⅜ as a decimal and a percentage.
3. If P(win) = 2/7, write the odds in favour as a ratio.
4. Spiral review: 13x - 12 = 5x + 60. Find x.
5. Spiral review: Area of a semicircle with r = 7 (π = 22/7).

Exercise 3 — Without replacement (includes mock)

1. [Mock 2026 P2 Q5(a)(iii)] If 2 pens are picked in succession without replacement, what is the probability that both are green?
2. Bag: A, A, B. Pick two without replacement. Find P(both A).
3. In the pen bag (5 red, 3 blue, 2 green), find P(red then blue) without replacement.
4. Find P(blue then red) without replacement.
5. Spiral review: If the area of a circle is 64π, find the circumference.

Exercise 4 — Sample spaces

1. List the sample space for two coin tosses.
2. List the sample space for one die roll.
3. P(rolling a 7 on a die) = ?
4. P(a day has 24 hours) = ?
5. Spiral review: Classify −3 under ℕ, 𝕎, ℤ, ℚ.

Exercise 5 — Complements and "not" events

1. In the pen bag, find P(not green).
2. P(not red) = ?
3. P(not an ace) = ? (from a standard deck)
4. Spiral review: Solve 2x + 1 > 9.
5. Spiral review: (−3)^(x-2) = −243. Find x.

Exercise 6 — Two-stage independent

1. Two coins: find P(both heads).
2. Two coins: find P(two tails).
3. Two coins: find P(one head, one tail) in any order.
4. Spiral review: A van travels 154 km in 1 ¾ h. Speed?
5. Spiral review: n/100 = 10.5. Find n.

Exercise 7 — Real-life probability

1. A class has 20 pupils, 8 girls. Find P(girl) as a fraction, decimal, and percentage.
2. A bag has 6 blue and 4 red marbles. Find P(blue).
3. A spinner has 4 equal sectors, 1 red. Find P(red).
4. Spiral review: Find the mean of 3, 5, 7, 9.
5. Spiral review: Median of 20 marks from the stem-and-leaf (57.5) — verify the two middle values.

Exercise 8 — Challenge

1. Die thrown twice. Find P(at least one six).
2. In the pen bag, find P(two different colours) when two pens are drawn without replacement.
3. A card is drawn from a standard deck. Find P(ace or king).
4. Spiral review: In a market of 36 fishmongers, 28 sell mackerel, 18 tilapia, 13 both. How many sell neither?
5. Spiral review: Find the gradient through (1, 2) and (4, −4).

9. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. 5/10 = ½.
2. 3/10.
3. 2/10 = ⅕.
4. ½.
5. x = 3 → C.

Solutions to Exercise 2

1. (α) 0.3; (β) 30%; (γ) 3 : 7 (or 3 : 10).
2. 0.375 = 37.5%.
3. 2 : 5.
4. x = 9.
5. 77 cm^2.

Solutions to Exercise 3

1. 2/10 × 1/9 = 1/45.
2. ⅔ × ½ = ⅓.
3. 5/10 × 3/9 = ⅙.
4. 3/10 × 5/9 = ⅙.
5. 16π.

Solutions to Exercise 4

1. {HH, HT, TH, TT}.
2. {1, 2, 3, 4, 5, 6}.
3. 0.
4. 1.
5. $\mathbb{Z}$, $\mathbb{Q}$.

Solutions to Exercise 5

1. 8/10 = ⅘.
2. ½.
3. 48/52 = 12/13.
4. x > 4.
5. x = 7.

Solutions to Exercise 6

1. ¼.
2. ¼.
3. 2/4 = ½.
4. 88 km/h.

5. n = 1050.

Solutions to Exercise 7

1. ⅖ = 0.4 = 40%.
2. 6/10 = ⅗.
3. ¼.
4. 6.
5. 10th value 57, 11th value 58; median = 57.5 (correct).

Solutions to Exercise 8

1. 1 - ≤ft(⅚)^(2) = 11/36.
2. 1 - P(both same) = 1 - ≤ft(1/45 + 3/90 + 3/90) = 10/15 = ⅔ (see note).

Detail: P(GG)=1/45, P(RR)= (5/10)(4/9)=2/9, P(BB)=(3/10)(2/9)=1/15. Sum same-colour = 1/45+2/9+1/15 = 1/45+10/45+3/45 = 14/45. Different colour = 1 − 14/45 = 31/45.

3. 4/52 + 4/52 = 8/52 = 2/13.
4. 3.
5. m = −2.

10. Authentic Mock Items in This Lesson

Ref	Question	Answer
P2 Q5(a)(i)	P(red)	½
P2 Q5(a)(ii)	P(blue) as decimal, %, ratio	0.3 = 30% = 3:7
P2 Q5(a)(iii)	Both green, without replacement	1/45

11. Lesson Summary

Skill	Rule
Basic probability	favourable ÷ total
Forms	fraction → decimal → % → ratio
Without replacement	reduce total and remaining favourable
Complement	P(not A) = 1 - P(A)
Two-stage (independent)	multiply
Two-stage (dependent)	multiply with updated numbers

Lesson prepared under `Grade9-Maths/Rules.md` — plain English, British notation, full workings required.